Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table? 2

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? Yes 2

Does R2 show up as an OSPF neighbor on R3? No 2

What does this information tell you?

h. R2 has enabled OSPF advertising only on interface S0/0/0 and establishes a neighbor relationship only with R1; R2's S0/0/1 interface remains passive and cannot exchange Hello packets, so there is no neighbor relationship. Consequently, R3 can access the 192.168.2.0/24 network only via the R1→R2 hop 2

Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1_no passive-interface s0/0/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network?

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost is 65. It is calculated by adding the cost of the serial link 64 fromR3 to R2andtheGigabitEthernet interface cost 1 on R2.

2

Is R2 listed as an OSPF neighbor to R3? Yes 2

Change OSPF Metrics

Step 2: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default OSPF reference bandwidth is 100 Mbps, which causes all links faster than 100 Mbps tohavethe same cost of 1. Changing the reference bandwidth allows OSPF to more accurately reflect the differencesin link speeds such as Gigabit and 10 Gigabit Ethernet 2

Step 3: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_OSPF adds the cost of each interface along the path. The total cost to the network is the sumof all interface costs from the source router to the destination. 2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562 because the bandwidth of the serial interfaces was changed to 128 Kb/s. Thisincreased the OSPF cost of the interfaces, so the total path cost became higher. 2

Step 4: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The route goes through R2 because the cost of the link to R3 was increased using the ip ospf cost command.OSPF chooses the path with the lowest cost, so R1 now uses the route through R2. 4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   R2 has enabled OSPF advertising only on interface S0/0/0 and establishes a neighbor relationship only with R1; R2's S0/0/1 interface remains passive and cannot exchange Hello packets, so there is no neighbor relationship. Consequently, R3 can access the 192.168.2.0/24 network only via the R1→R2 hop.

2. Why is the DR/BDR election process not a concern in this lab?

   The OSPF Router ID is a unique identifier for routers within an autonomous system, used in core processes such as establishing adjacencies, generating and exchanging LSAs, and electing DRs and BDRs; Unless manually configured, the router automatically selects the IP address with the highest activity on a physical interface as the Router ID. If that interface fails or the IP address changes, the Router ID will change, causing OSPF adjacencies to be disrupted, LSAs to be retransmitted, routes to converge, and potentially triggering routing loops;

3. Why would you want to set an OSPF interface to passive?

   The purpose of a passive interface is to prevent the interface from sending or receiving OSPF Hello packets while still allowing the network segment associated with that interface to be advertised in OSPF. Key benefits include:

   Reduced network traffic: LAN interfaces (such as G0/0, which connects to PCs) have no other OSPF routers, so there is no need to exchange Hello packets, thereby reducing link bandwidth consumption;

   Enhanced network security: Prevents OSPF protocol messages from being exposed on non-routing segments, thereby preventing malicious devices from forging OSPF adjacencies or tampering with routing information;

   Simplified OSPF configuration: Eliminates the need to configure additional OSPF parameters for interfaces with no OSPF neighbors, reducing the overhead of maintaining adjacency relationships.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF's area partitioning mechanism is central to its hierarchical design and support for large-scale networks. OSPF divides an autonomous system (AS) into multiple areas, which must include a backbone area (Area 0); all non-backbone areas are directly connected to the backbone area.

   Routers within an area maintain only the topology information for that area and calculate intra-area routes using the SPF algorithm.

   Routing information between areas is aggregated and transmitted via Area Border Routers (ABRs), preventing LSAs from flooding the entire AS. This hierarchical design reduces the size of each router's topology table and routing table, lowers CPU and memory overhead, and accelerates routing convergence, enabling OSPF to support large-scale networks with thousands of routers

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0 The OSPF `network` command uses ** **wildcard mask** to match interface IP addresses. 0.255.255.255 indicates that "the first octet is fixed at 10, and the remaining three octets can be any value." This matches all interfaces in the 10.0.0.0/8 network segment and assigns them to Area 0.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

When an EIGRP router selects a route, **Administrative Distance (AD)** takes precedence over the metric: the lower the AD, the higher the protocol's reliability. Default Cisco IOS administrative distances:

4

RIP: 120

OSPF: 110

EIGRP (internal): 90

Among these three, EIGRP has the lowest AD, so the router will select the EIGRP route regardless of its metric value.

I can help you organize a list of core configuration commands for this lab (categorized by R1/R2/R3, including basic configuration, OSPF configuration, passive interfaces, and metric modification). Would you like that?